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Beyond $\alpha = 0.05$

A Data-Driven System for Alpha Level Selection

Master's Thesis

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Hypothesis Testing

Observed Result

- "A method shows improvement"

Key Question

Is it **real** or just **coincidence**?

$\Downarrow$

Hypothesis Testing Procedure

- Assume H_0: No effect
- Choose α (risk level)
- Compute p-value (under H_0)
- Compare p-value with α

Decision Rule

Decision based on p-value and α

$$p \leq \alpha \;\Rightarrow\; \text{Reject } H_0$$

$$p > \alpha \;\Rightarrow\; \text{Do not reject } H_0$$

α determines what is considered "surprising"

The Problem with $\alpha = 0.05$

- $\alpha = 0.05$ become standard threshold with no clear justification

One Size Fits All

Same α used regardless of

- sample size
- effect size

Cliff-Edge Effect

- $p = 0.049 \rightarrow$ Significant
- $p = 0.051 \rightarrow$ Not significant

Replication Crisis

- Many statistically significant findings fail to replicate.

[Ope15; WL16]

Existing Approaches for Choosing α

- Fixed α levels (0.10, 0.05, 0.01)
- Power-based approaches (sample size, effect size)
- Decision-theoretic approaches (error costs)
- Stricter thresholds ($\alpha = 0.005$)

$\Longrightarrow$ **All approaches still rely on a fixed α**

[Coh88; Ber85; Ben+18]

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Proposed Alpha Recommendation System

Goal: Recommend α based on data & study context

INPUT		STAGE 1		STAGE 2		OUTPUT
Sample Size + Effect Size (Known initially)	$\rightarrow$	**Random Forest Regression** Baseline α Learned from previous data	$\rightarrow$	**Context Adjustment** (K1) (k2) (k3)	$\rightarrow$	**Final α Recom-mendati-on** Clamped to 0.001 – 0.15

Data Source

- Replication studies (OSF)
- Across multiple research fields
- 664 hypothesis tests
- Includes:
  - Sample size
  - Effect size
  - p-values

[Cha25; ED21; Sot17; Sot22; Li21; Ope15]

Example Dataset Rows

Sample Size	Effect Size	p-value	Optimal α
120	0.35	0.041	0.052
85	0.28	0.063	0.061
241	0.30	0.048	0.074
150	0.22	0.071	0.066
96	0.40	0.032	0.049

- Each row represents one hypothesis test.

Optimal Alpha Definition

Replicated

Keep original α
(decision validated)

Failed

Adjust α: largest
$\alpha < p$-value
(correct decision)

Example

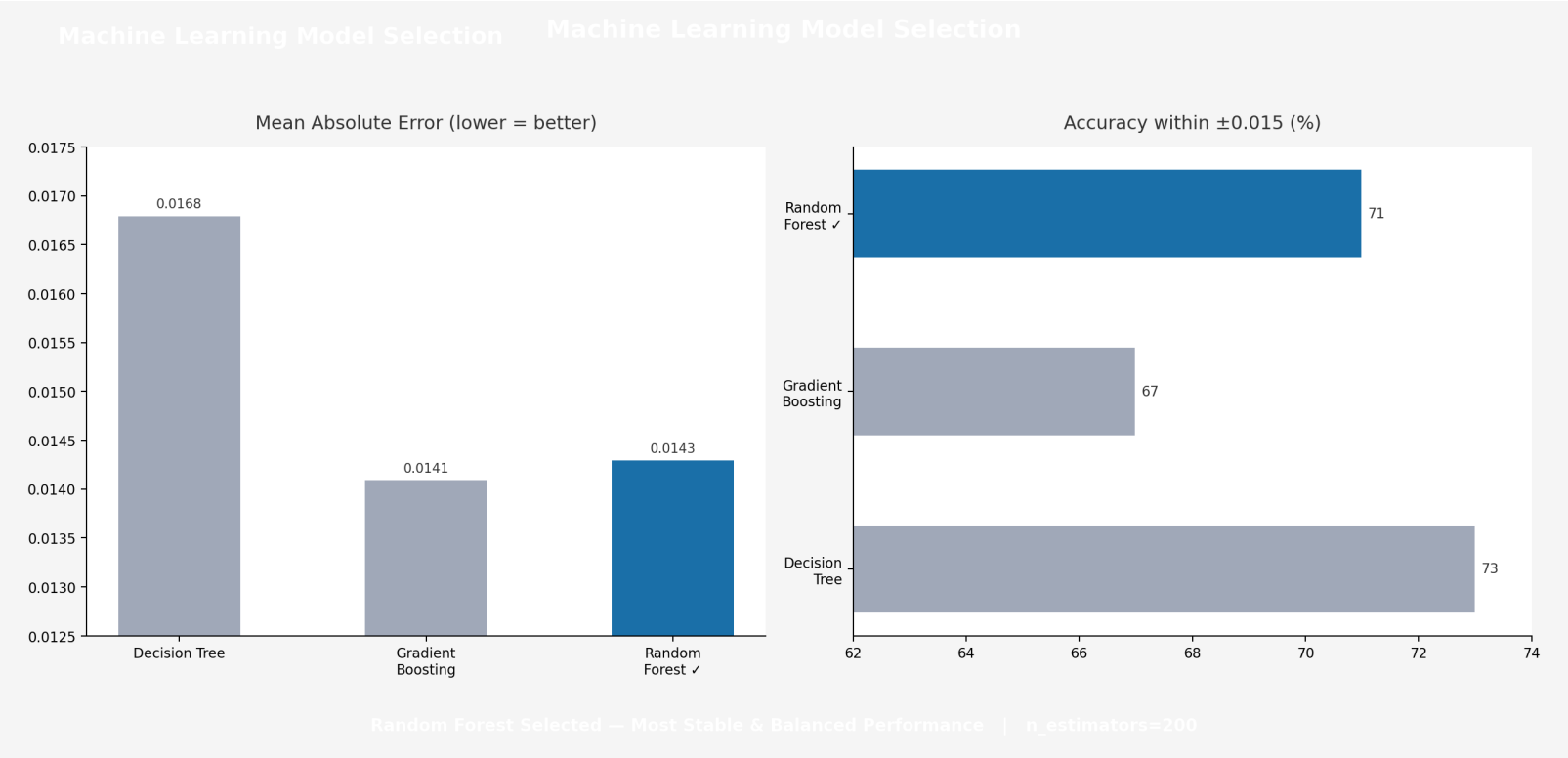
$p = 0.047 \rightarrow$ failed to replicate
Optimal $\alpha = 0.04$ (not 0.05)

Predicting Baseline α

Machine Learning Task

- Goal: predict α_{baseline}.
- Input features:
  - Sample size
  - Effect size
- Target variable: optimal α .

Machine Learning Model Selection



Random Forest achieves the lowest error

Selected Model: Random Forest

- Lowest prediction error
- Handles non-linear relationships.
- Stable performance

Implementation

Random Forest with **200 trees** used for baseline α prediction.

Stage 2: Context Based Adjustment

Baseline α is not sufficient

Reason

- Same inputs can have different study contexts
- Baseline α does not reflect user preferences or risk
- Real world decisions require context aware adjustment

Adjustment factors

k_1 — Risk Tolerance

- Very Strict $\times 0.50$
- Strict $\times 0.75$
- Neutral $\times 1.00$
- Lenient $\times 1.25$
- Very Lenient $\times 1.50$

k_2 — Test Direction

- Two-tailed $\times 1.00$
- One-tailed $\times 2.00$

k_3 — Multiple Testing

Bonferroni:

$$\alpha/m$$

$$\alpha_{final} = \alpha_{baseline} \times k_1 \times k_2 \times k_3$$

Final range: $0.001 \leq \alpha_{final} \leq 0.15$

System Deployment

Implementation

- Implemented in **Python**.
- ML model:**scikit-learn**.
- interface:**Streamlit**.

$$\text{User Inputs} \implies \text{Recommended } \alpha$$

System Interface

Evaluation Setup

Dataset

- Tested on 20 failed replication studies

Comparison

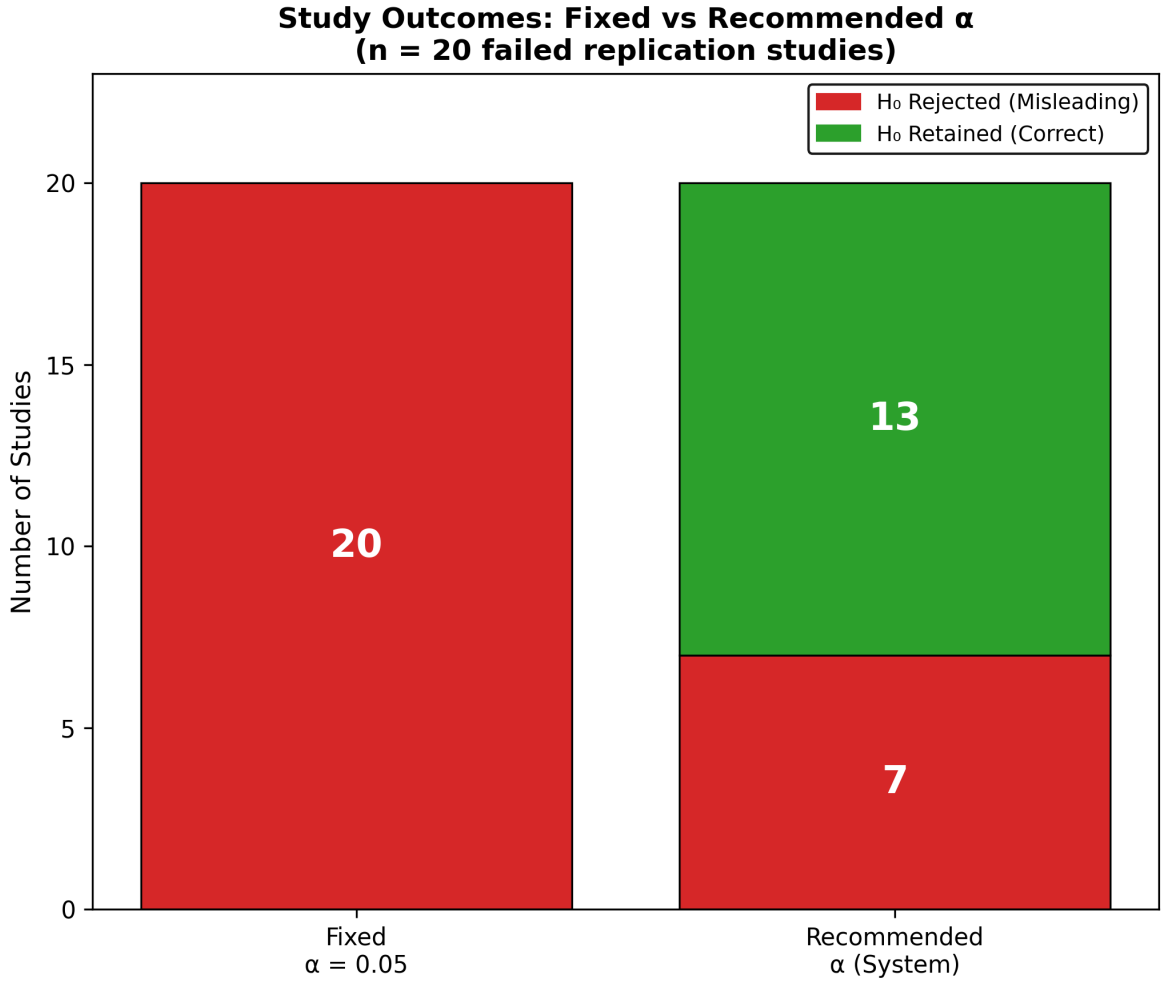
- $\alpha = 0.05$
- Recommended α

Evaluation

- Decision change (reject $\rightarrow$ not reject)

$\Longrightarrow$ **Goal: Avoid false positives**

Main Findings



Reduces false positives by 65% (13 out of 20 cases)

Limitations

Key Limitations

- Limited dataset size
- Heuristic adjustment factors
- Neutral context used in evaluation
- Results provide guidance, not strict rules

Conclusion

Key Takeaways

- $\alpha = 0.05$ is not universally appropriate
- α should depend on study characteristics
- A data-driven approach improves α selection
- Combines machine learning with interpretable adjustments
- **Statistical significance should be adaptive, not fixed**

Future Work

Next Steps

- Expand replication dataset
- Include additional study features
- Explore alternative ML models
- Evaluate beyond only rejected cases
- Deploy as a fully accessible web application

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Vielen Dank

für Ihre Aufmerksamkeit

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